

Name: _____

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MyDesign Portfolio

Element B Initial Template - AT - 6 points

Element B: Documentation and Analysis of Current and Prior Design Attempts

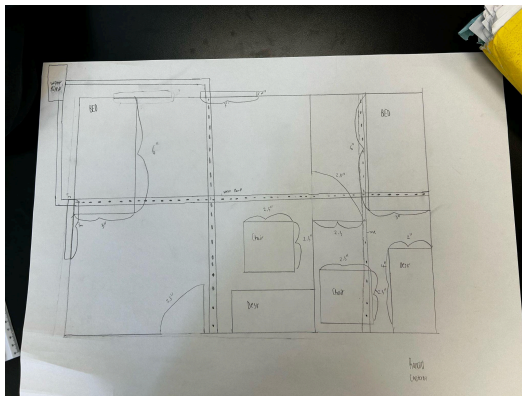
Current or Prior Design Attempt #1

Source

Our own group's final hand-drawn sketch of our prototype. It is based on UMD's Burn Day design requirements, and we drew our fire safety systems on the sketch.

<https://fpe.umd.edu/sites/fpe.umd.edu/files/FPE-Design-Challenge-MANUAL.pdf>

Image



Details

This design attempt is a sketch of our room, and it represents our early vision for our fire safety system. It includes the room's measurements, furniture placements, and fire safety system. This version of our design reflects our team's initial efforts and ideas to contribute to our future final design.

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Analysis

The sketch barely scratched the surface of what we were building, although it was to scale, it didn't help us envision our future fire safety system prototype. The sketch lacked accurate dimensions, so it wasn't to scale with what we actually built. We couldn't envision what our prototype would look like until we started building it.

Prior Design Attempt #2

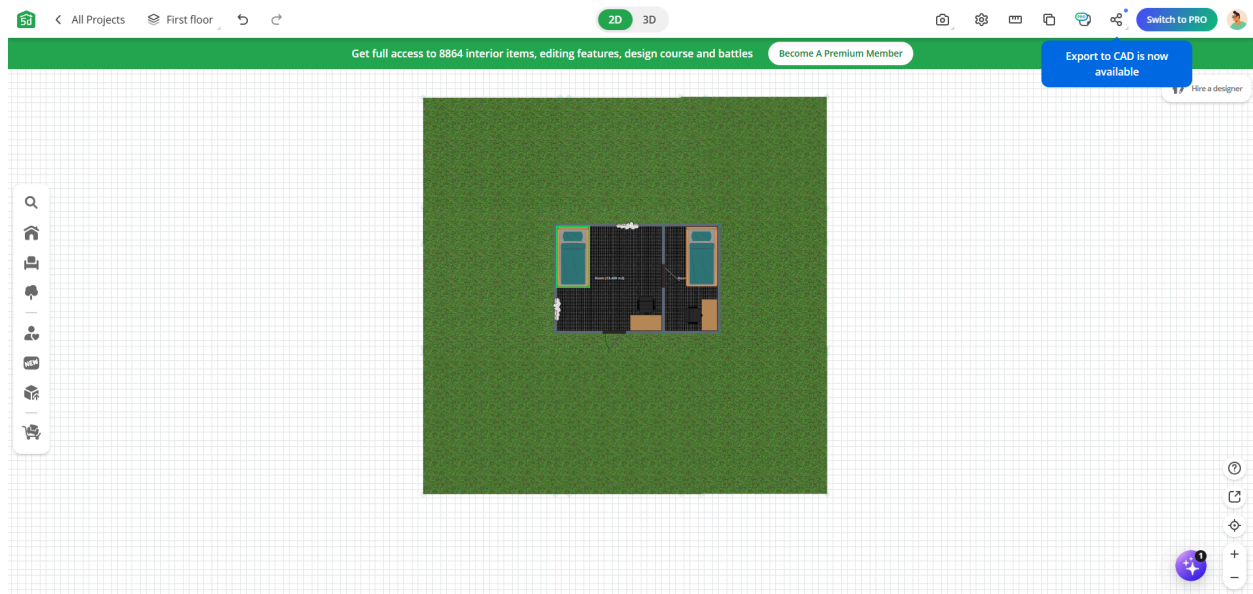
Source

Our digital sketch was created using Planner5D; we pretty much put our design dimensions into it and recreated our sketch in a digital format. We also kept using our design manual.

<https://fpe.umd.edu/sites/fpe.umd.edu/files/FPE-Design-Challenge-MANUAL.pdf>

<https://planner5d.com/>

Image



Details

This version of our sketch helped us envision our build, and we could see our sketch in 3D. It allowed us to imagine the space we were going to be working with, especially with our furniture

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we are supposed to have inside our room design. We also made sure to keep following the UMD Burn Day Manual that I mentioned earlier.

Analysis

Even though this version of our design was a lot more accurate compared to our initial sketch, it was limited by a lack of real-world testing. We couldn't test out the materials, and we couldn't implement how our fire safety system would work. We still couldn't envision our actual fire safety system and how it would work.

Prior Design Attempt #3 - Cardboard house

Source

We built this cardboard prototype off of the rubric that was given to us by UMD.
<https://fpe.umd.edu/sites/fpe.umd.edu/files/FPE-Design-Challenge-MANUAL.pdf>

Image



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This prototype was made with cardboard to test the space and create furniture, and see how much room we will have to work with when we install our suppression system. This prototype allowed us to envision our fire safety system.

Analysis

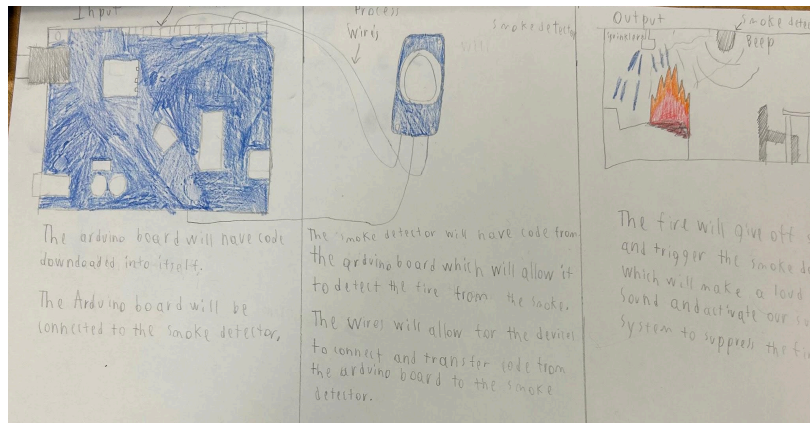
The cardboard model was useful to us because it allowed us to envision our space, but we still couldn't test anything with it. We couldn't see how our fire safety system would've interacted with the space, and the model was also uneven.

Prior Design Attempt #4

Source

This source allowed us to develop our different systems for our fire safety system. It allowed us to understand.

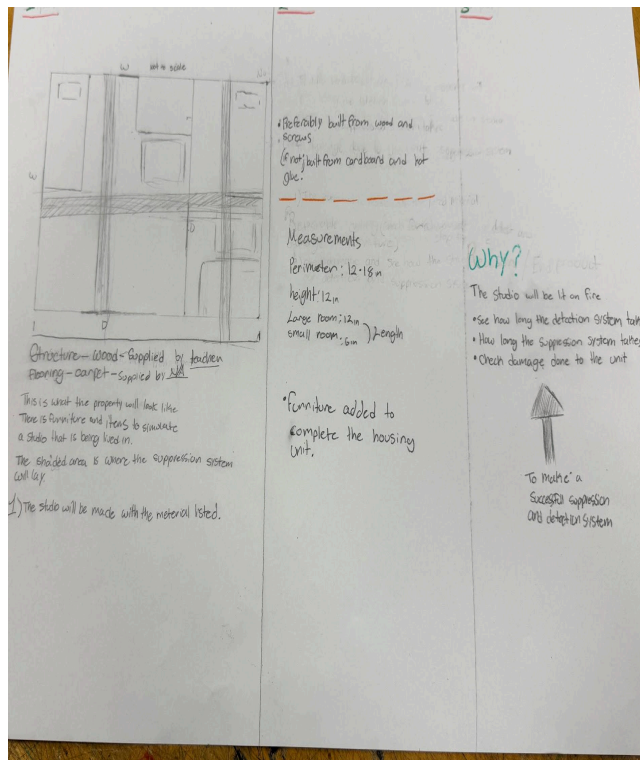
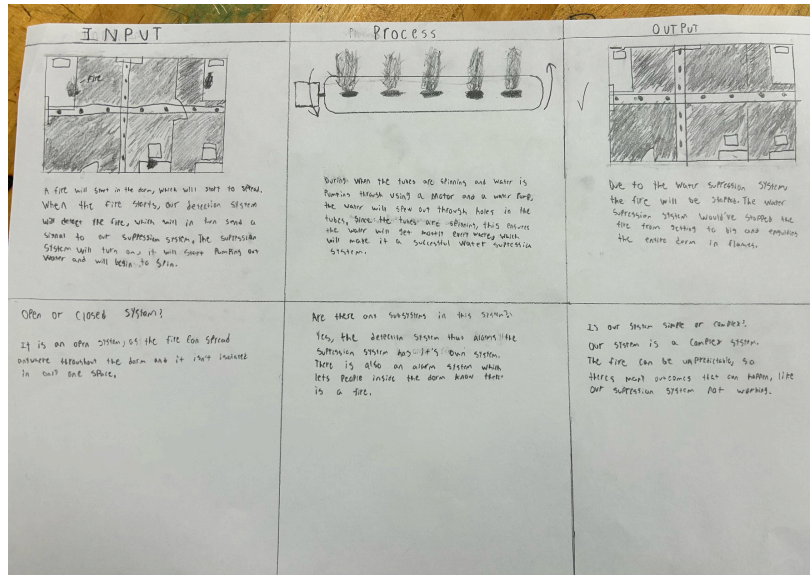
Image



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These systems of our design for our fire system safety allowed us to understand how a system works. We focused on the interactions between our different systems and how it would work together. We envisioned our subsystems, like our detection system, suppression system, and the design of our room.

Analysis

While these forms helped us understand our systems, we still couldn't interact with our systems or the design of our room. It wasn't physical, it was just something we used to comprehend our ideas.

Prior Design Attempt #5

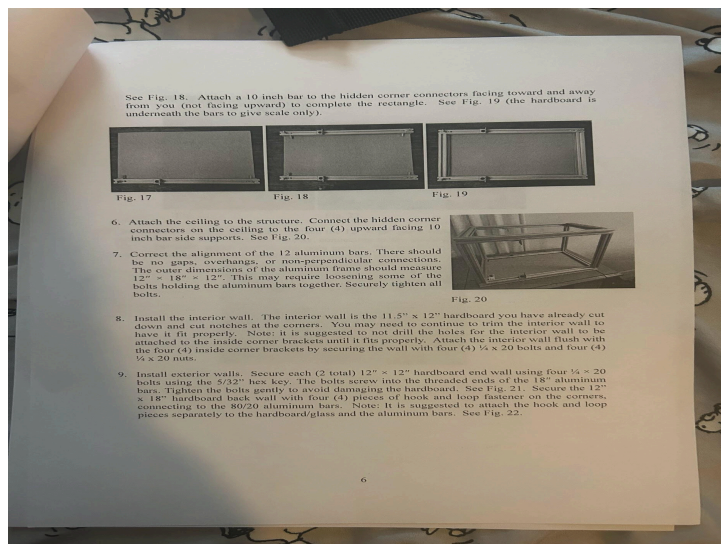
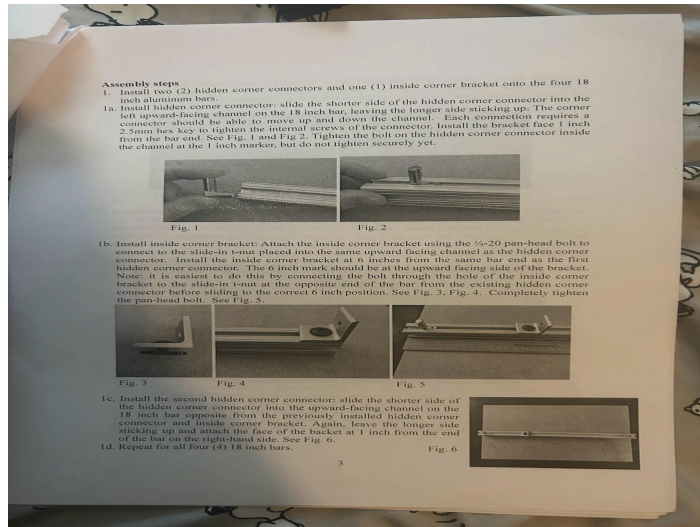
Source

For building our room, we followed these instructions in order to conform to the UMD Manual.

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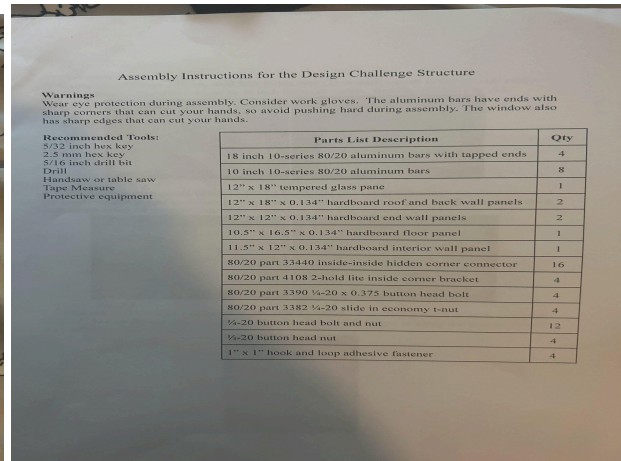
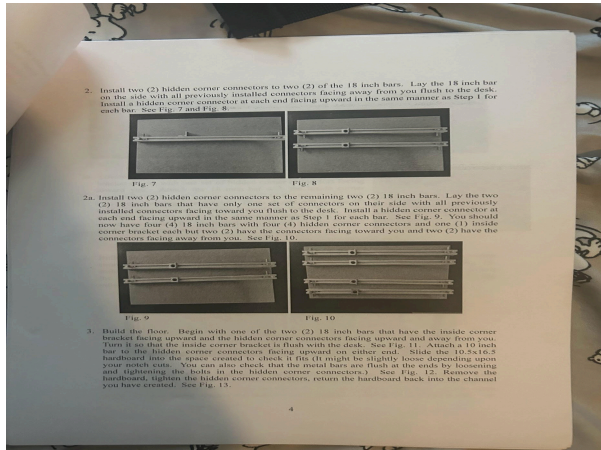
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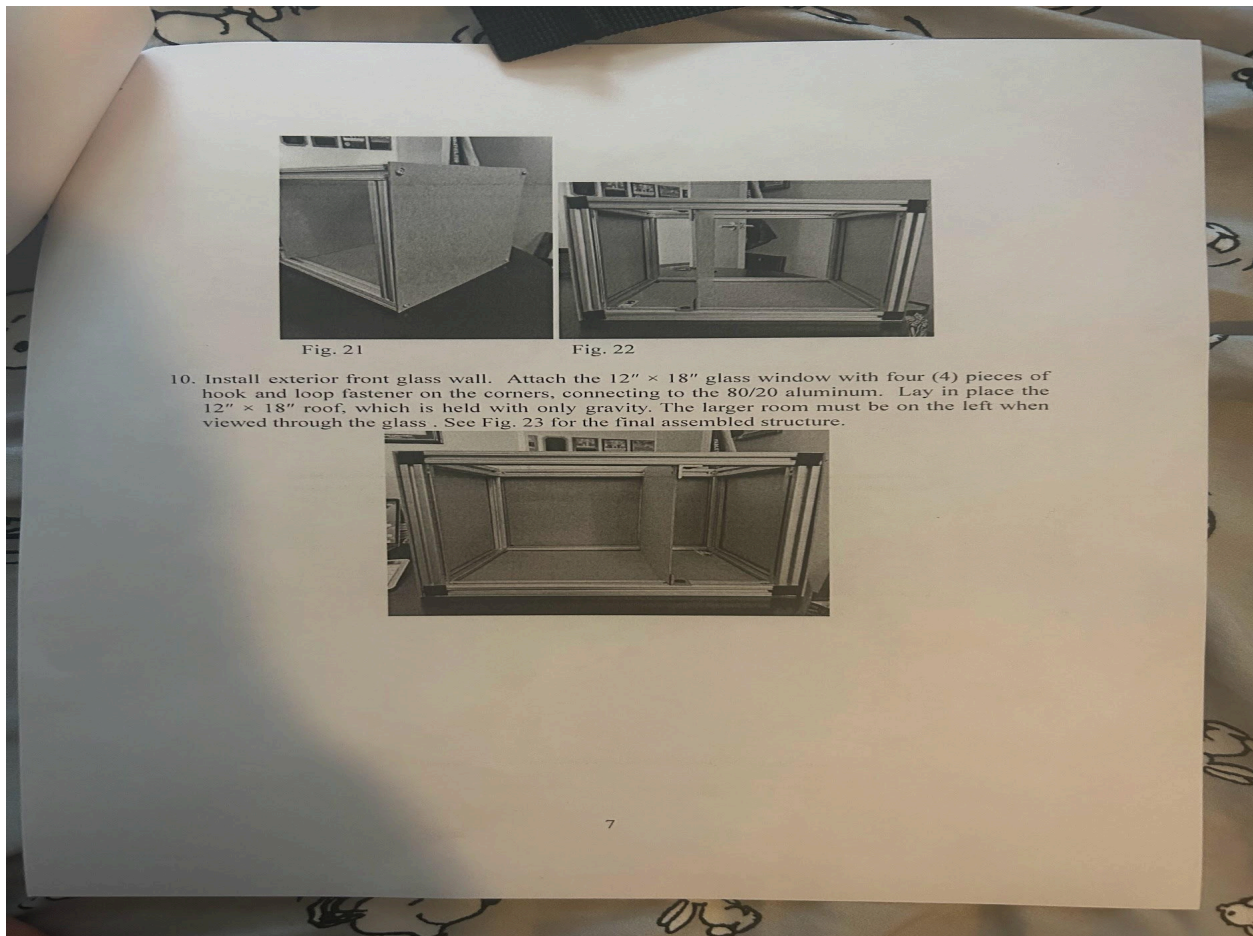
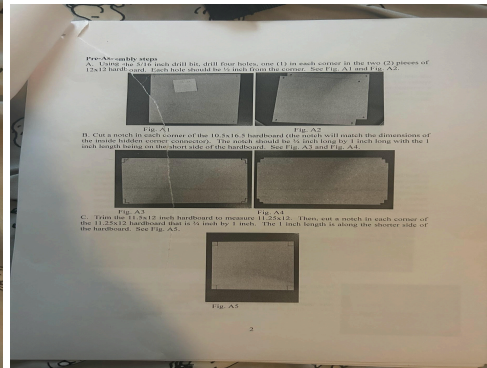
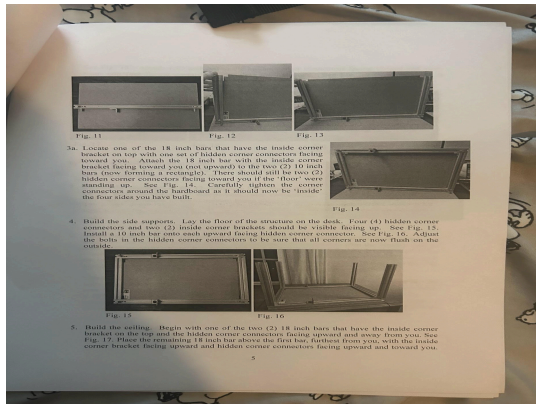
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Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).



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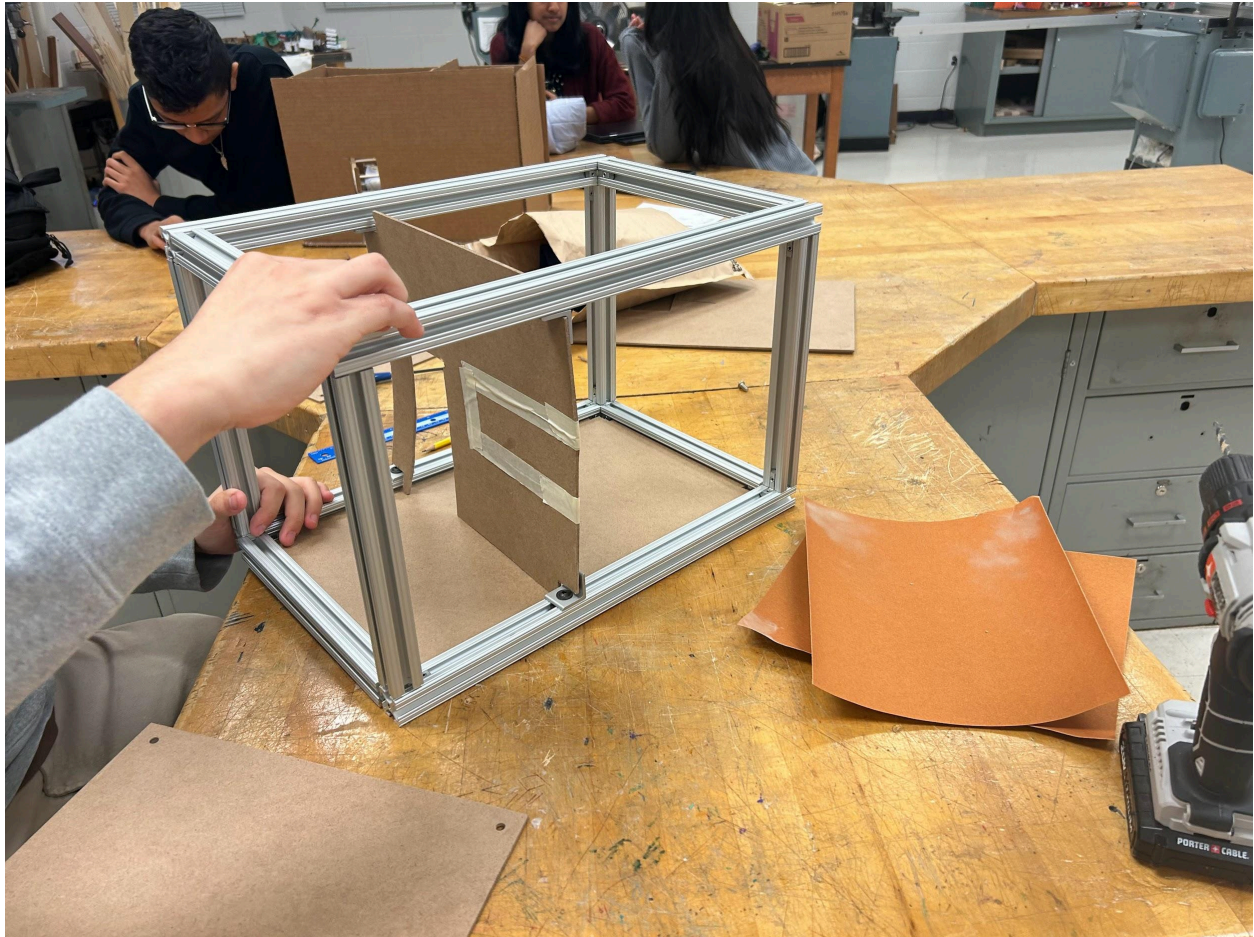
Image



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Details

Our final design is what we were working on for the entire 2 months we were working on the project. Our design had a very successful design system, and all of our furniture conformed to the UMD Manual. Our detection system worked; we used a coding device called Arduino, which was a device we could put code inside of and connect it to devices that do different things. It was successful in detecting the fire and alerting to make sure that, if people were inside, they would be able to know something is wrong.

Analysis

Our prototype was unsuccessful because it couldn't put out the fire that it was supposed to. We also weren't able to have a complete build. The fire, unfortunately, was able to burn the

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furniture. Our final design lacked a suppression system that was unable to put out the fire inside of our design.

Rubric

Element B: Documentation and Analysis of Prior Design Attempts						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B1. Documentation of plausible prior attempts to solve the problem and/or related problems is drawn from _____.	a wide array of clearly identified and consistently-credible sources	a variety of clearly identified and consistently-credible sources	several—but not necessarily varied—clearly identified and generally-credible sources	a limited number of sources, some of which may not be clearly identified and/or credible	only one or two sources that may not be clearly identified and/or credible	missing or minimal (a single source that is not clearly identified and/or credible)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B2. The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — _____.	is consistently clear, detailed, and supported by relevant data	is clear, generally detailed, and supported by relevant data	is generally clear and contains some detail and relevant supporting data	is overly general and contains little detail and/or relevant supporting data	is vague and is missing any relevant details and/or relevant supporting data	cannot be inferred from information intended as analysis of past and/or current attempts to solve the problem

